SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.* (BL2, BL3)

QUESTIONS: 1

Total Marks:100

Annexure A

Concept Mapping

Code of Conduct:

I M. Venkatesh 192511133 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Venkatesh

Annexure A

Concept Mapping

Question 1
Show a multi-level concept map integrating:
Operating System Types → OS Structures → System Calls → OS Functions.
Your map must show cross-links (e.g., how OS structure influences system call implementation). **(BL2)**

Question 2
Construct a comprehensive concept map representing:
Process Management, Memory Management, File System, and I/O System.
Include interdependencies and indicate where system calls interact with each component. **(BL3)**

Question 3
Sketch a comparative concept map for:
Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.
Highlight differences, similarities, advantages, and limitations using linking phrases.
(BL2)

Question 4
Illustrate a dynamic concept map showing the execution flow:
User Program → API → System Call Interface → Kernel → Hardware → Response back to User.
Include feedback loops and control flow paths. **(BL3)**

Question 5
Sketch a hierarchical + functional concept map for system calls:
Process, File, Device, and Information system calls.
Also show how they relate to OS services like scheduling, memory allocation, and device handling. **(BL3)**

RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

CSA04
operating systems

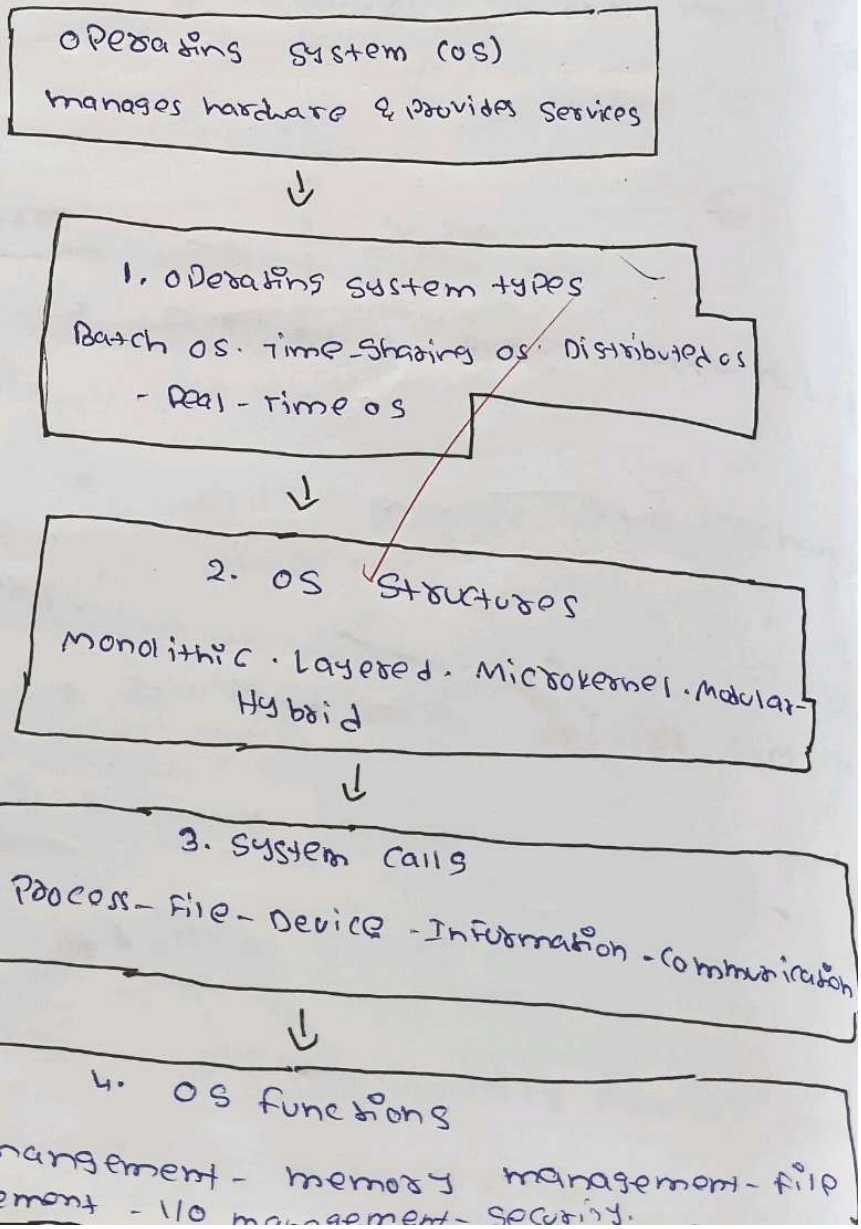
Assessment Tool - 2

Concept mapping

1)

Concept map integrating OS types, OS structures, system calls, & OS functions.

An operating system (OS) is system software that manages computer hardware and provides services to application programs & users. Different types of operating systems use different structures system calls and functions according to their needs.



1) Operating system types

- Batch os: Executes a group of jobs without continuous user interaction
- Time-sharing os: Allows multiple users or programs to share CPU time
- Distributed os: Coordinates resources across multiple connected computers.
- Real time os: Produces responses within specified time limits.

2) os structures

- Monolithic structure: Most operating system services run inside one large kernel.
- Layered structure: The os is divided into layers with each layer using services from lower layers
- Microkernel structure: only essential services run in the kernel; other services run separately
- Modular structure: The kernel uses separate modules that can be added or removed.
- Hybrid structure: Combines features from different os structures.

3) System calls

- Process calls: create, execute & terminate processes
- File calls: Create, open, read, write, & close files.
- Device calls: request access to hardware devices.
- Communication calls: Exchange data b/w processes or computers.

a) OS Functions

- Process management: Creates processes schedules CPU time & handles termination
- memory management: Allocates & retrieves memory
- File management: organizes files & directories
- I/O management: controls input & output devices
- security & protection: controls access to resources.

1) OS type -> OS Functions: The type of OS determines its priorities. For example a real-time OS emphasizes predictable response times.

2) OS structure -> System calls: The kernel structure determines how requests reach kernal services & how those are organized.

3) System calls -> OS Functions: System calls allow programs to request operations such as file access and process creation.

4) OS functions -> Hardware: The OS manages CPU memory storage, & devices.

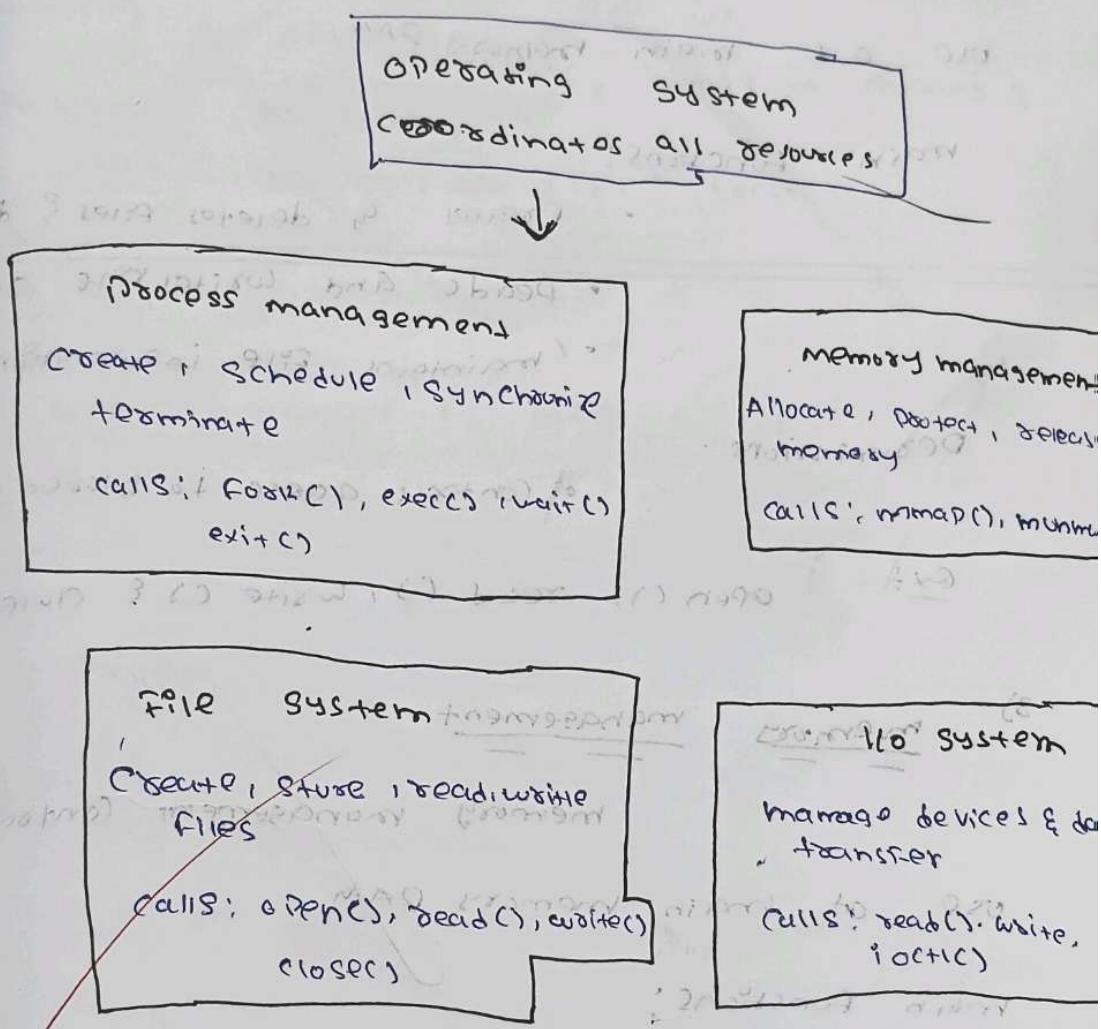
5) OS. Structure -> security & reliability: The separation of services and kernal components affects fault isolation and protection.

1)

Concept map for process management memory management, file system, & I/O system

A)

An operating system manages processes, memory, files, & input output devices. These components depend on one another to execute programs efficiently.



1) Process management

A process is a program that is currently running

main functions:

- creates & terminates processes
- schedules processes for CPU execu

- Handles waiting and process states.

Ex: Fork(), exec(), wait(), exit()

2. ~~Memory management~~. File System

Memory management controls the use of main memory RAM.

main functions:

- Creates & deletes files & directories
- Reads and writes file contents
- maintains file information & permissions.
- Controls access to stored data.

Ex: open(), read(), write() & close()

3) Memory management

Memory management controls the use of main memory RAM

main functions:

- Allocates memory to processes
- Releases memory when it is no longer required
- protects the memory space of each process
- Supports virtual memory.

Ex: mmap() & munmap() on unix-like systems

4)

I/O system

The input/output system manages communication b/w the computer & its devices.

main functions:

- controls devices through device drivers.
- Transfers data b/w devices & memory
- Supports buffering caching & interrupt handling.

Ex:- read(), write(), & ioctl().